



Society of Petroleum Engineers

SPE-229089-MS

Digital Gas Management Solution for Sustainable Operations

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/229089-MS](https://doi.org/10.2118/229089-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

In the Amazon region having cleaner and sustainable operations is a key factor to develop a brown field, looking for the continuous improvement of operations, reaching net zero goals and complying with local and international legal regulations while reducing cost and optimizing production. Therefore, with the commitment of eliminating all the well pad flares, this project presents an integrated digital solution using a compressor, valves, sensors and instrumentation, advanced analytics, edge and cloud computing, and Internet of Things (IoT) technologies to close well pad flares, ensuring artificially lifted wells stability, avoiding gas handling problems and optimizing production while significantly reducing the carbon footprint generated.

By installing a compressor to manage annular well gas from all the wells of a platform, the Operator can close the flare to comply with local legal regulations and send the gas through the production pipeline to the flow station and to avoid Electro Submersible Pumps (ESP) in the wells having gas handling problems and instability due to this operation. This solution integrates the use of edge and cloud advanced workflows connected to a Supervisory Control and Data Acquisition system (SCADA) that using IoT, remotely and autonomously control annular valves to generate well stability and to optimize pump intake pressure (PIP) to enhance production based on the ESP and surface sensor data gathered, ensuring the best submersion point to operate the artificial lift equipment. All this process is monitored in a collaborative visualization environment considering the high and low frequency data generated from the solution and giving alerts according to the operational events.

This Smart Production Operation (SPO) Gas Management solution implemented in a remote field in the Amazon has shown remarkable results for digital value creation, improving production and reducing related losses in 130 BOPD, due to the gas handling events avoided. This solution also ensures well stability, reducing ESP anomalous conditions, enhancing equipment run life (RL), but most significantly reducing the carbon dioxide equivalent (CO_{2e}) emissions in 9 kton per well pad per year. In addition to these benefits, the Operator used to optimize or debottleneck the gas handling issues manually well by well and now all

the optimizations are done remote and autonomously, reducing commuting time to the well pad, and Health Safety and Environment (HSE) events.

The SPO Gas Management solution not only eliminates gas flaring but also with the novel and unique implementation of advanced analytics with IoT for remote and autonomous control, reduces manual operations and commuting time to well pads, promoting cleaner production and operations, minimizing HSE-related events, listening to the community needs of reducing or eliminating flares in the Amazon region, supporting the protection of the planet's most diverse ecosystem and generating better perspective for the energy industry.

Introduction

In the last years, the Ecuadorian oil industry has been pushed to have more sustainable operations due to its social and environmental impact, especially in flaring activities. The legal context in Ecuador in which now Oil and Gas companies are required to operate under, include the total removal of flares scattered around the Amazon Region, due to the impacts caused to plantations, environment and the surrounding communities (Diantini et al., 2020). Therefore, the operators must find a technical solution to avoid impacts in production with no-flaring conditions (Fig. 1).



Figure 1—Flares in a wellpad in the Ecuadorian Amazon Region.

This requirement applies to all the oil producing fields of the Amazon Basin, which in their majority are comprised of brown fields that have been in production with an average of 50 continuous years (Finer et al., 2015). In such a context, operations in this kind of fields are related to significant reservoir pressure depletion, as well as an increase in general gas oil ratio (GOR). This in kind means that there is an expectation that gas handling will become a more pressing challenge for operators in order to keep producing in mature fields, technical solutions that take into consideration associated gas will become critical for production continuity.

This project is implemented in a brown field where the predominant artificial lift system is ESP, where gas management is one of the primary considerations when designing and operating the wells to have stable production. The use of flares in this context reduces the accumulation of gas in the annular space, so that gas handling issues within the ESP are mitigated, thus helping to reach the highest possible operating efficiency.

The current configuration at wellhead consists of a connection between the annular space through the wellhead's section B to the flowline with a check valve as a control device. This configuration at surface permits the movement of gas accumulated in the annular space to the flowline once pressure in the annular

space is at least equal to the flowline pressure, so that the check valve can open and permit this flow of gas as can be seen in (Fig. 2).

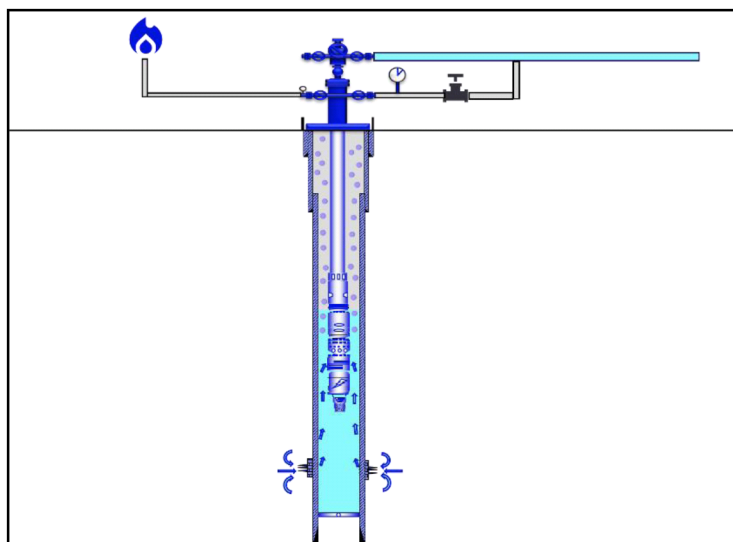


Figure 2—Wellhead configuration with annular-flowline connection.

While the configuration of the connection of the annular space to the flowline is sufficient for most wells to allow the management of gas accumulation, for those wells that have a considerably high GOR, this can no longer be managed by the current connections and leads to lifting problems, specifically ESP gas blockages and reduced efficiency. Therefore, the use of flares to lower the annular pressure is needed to reach stable production conditions, that at the start of 2024 represented 10 permanent and sporadic flares scattered around the field. That is the case and challenge of the well pad where this solution has been implemented, with 11 active wells and an average of 1,2 MMSCF per day and associated production of 5,600 BOPD, representing the highest producing well pad for the field.

The solution consists of an integrated gas management system in a SPO environment where a compressor is installed to handle the incoming gas from the annular space from all the wells in the well pad, and the annular pressure required by each well is in turn controlled by a set of automatic valves. This solution incorporates edge and cloud computing to determine the optimal annular pressure, and thus gas flow rate from the annular gas by utilizing an algorithm on a well-by-well basis integrating field data from surface instrumentation, pressure indicators and transmitters (PIT) and ESP operational parameters with a connection to the variable speed drive (VSD). The objective is to optimize gas flow from the annular space so that lift problems related to gas handling within the ESP are minimized therefore reducing production losses due to inefficiency and avoiding downtimes due to ESP blockages. Additionally, the solution has allowed production optimization by controlling ESP submersion while sending the previous flared on-site gas to a central processing station to be effectively managed or utilized.

Statement of Theory and Definitions

Flares in Oil & Gas operations

Historically, oil and gas operations have been associated with social and environmental impacts, however, the current paradigm is to find technological solutions to achieve more sustainable, efficient and less emissions heavy operations, thus complying with local and international regulations. One such technological challenge the industry has set its sights in is the reduction or the total replacement of flaring in operations.

A flare or flare stack is in essence a combustion device whose purpose is the safe disposal of excess or unwanted gas to the environment through combustion and is regulated by the API STD 521 standard (Hoque, 2013). This disposal of gas through burning is preferable to doing cold vents, or the release of the associated annular gas to the atmosphere, specifically in regards to emissions, as burning will convert by combustion the heavy concentration of methane gas (CH_4) into carbon dioxide (CO_2), thus reducing the greenhouse gas (GHG) potential of CH_4 which is up to 30 times as impactful as CO_2 (US EPA, 2016). In this case, the main purpose of flare use is the routine disposal of associated gas, which is considered to be uneconomical to capture and transport (U.S. Department of Energy, Office of Fossil Energy, 2019).

ESP and How Gas Affects Performance and Efficiency

ESP pumps are the most used artificial lift system in the area where the project is proposed. ESP pumps are composed of several components: a sensor to monitor downhole parameters, an electric motor powered from the surface via cable, protectors, intake, gas separators, and centrifugal pump bodies in series. Gas separator assemblies, which in this case mostly utilize vortex-type gas separators, separate the gas by sending it to annular space (Yanez et al., 2024). This separation efficiency is mostly impacted by gas accumulation in the annular space, which means that progressively less gas is being separated and thus it enters the pumping stages (Fig. 3), where although modern ESPs are designed to handle certain concentration of free gas, once gas fraction surpasses this level, current variations increase and lifting efficiency reduces (Awaid et al., 2014).

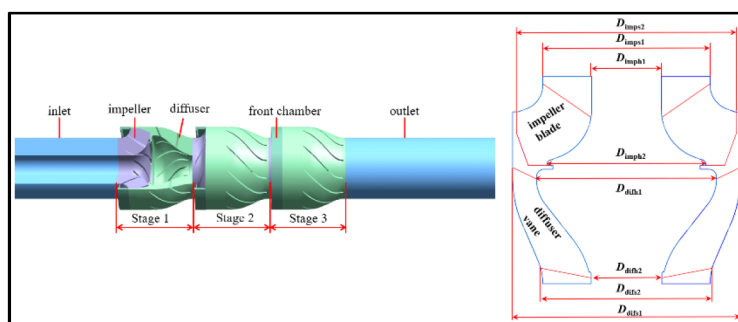


Figure 3—ESP impeller – diffuser configuration. D = diameter (Wang et al., 2023).

The impact of free gas on ESP pumps

- Reduction in the volume of fluid displaced: ESP stages are designed to move liquids, so the presence of gas reduces the volumetric capacity of each stage and consequently the lifting capacity, this effect is often described as gas blocking the ESP.
- Motor overheating: Free gas reduces the cooling capacity of the fluid surrounding the motor. The lack of adequate liquid flow at the bottom causes overheating and eventually a shutdown due to the activation of the high motor temperature trip alarm.
- Operational instability and equipment shutdown: The presence of gas is reflected in current fluctuations, which affect system stability. This effect can be as severe as fluctuating up to 33% of the normal operating current for the equipment, an effect directly correlated with constant unloading and loading of the motor and thus an increase in mechanical fatigue that can lead to premature mechanical failures of the equipment (Amijaya et al., 2020).

The above problems affect operations by increasing downtime losses and a reduced RL of equipment, leading to premature workover interventions for the equipment and thus deferring production.

Description and Application of Equipment and Processes

The implementation has two main components, hardware, referring to the equipment and instrumentation installed and adapted to the existing surface facilities; and software, referring to the connections of different data sources that feed the control algorithms performed by cloud and edge computing using physics-based correlations.

Hardware

One of the main components for the implementation of this project is a compressor system for handling the annular gas from the wells in the platform and carrying that gas to the production flowline. For this specific case, a reciprocal compressor system with a nominal capacity of 1 MMSCF per day is used to handle the total gas from the annular spaces for all 11 producing wells at the well pad. The normal operating conditions for this equipment are between 10 and 80 psig for the intake pressure and a discharge pressure between 60 and 165 psig. The volumetric gas flow for the equipment is given by the stroke rate of the piston, which in turn depends on the differential pressure between intake and discharge. Given that the discharge of the system will be connected to the flowline, this pressure gets equalized to the flowline pressure, meaning the gas rate that the equipment will handle will be in direct correlation with intake pressure. A diagram describing the general layout of the configuration for the compressor system is presented in (Fig. 4) with the equipment installed in (Fig. 5).

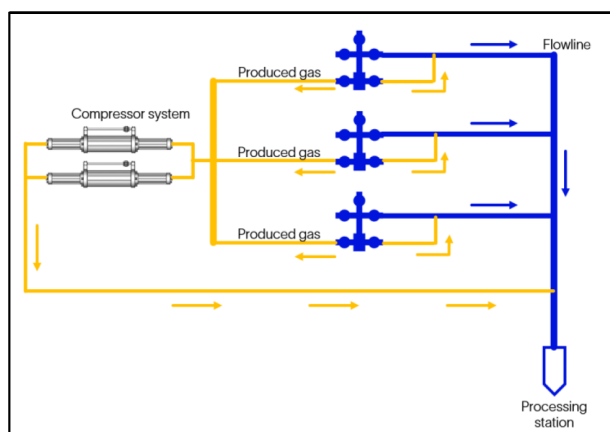


Figure 4—Schematic of the current implementation of the compressor system with annular gas at the well pad.



Figure 5—Reciprocating compressors used in this implementation.

The determination of the intake pressure for the compressor system will be given by the lowest working pressure, for one of the 11 wells. Meaning that the lower the intake pressure determined for a specific well, the higher the pressure differential for the compressor, and therefore the higher the stroke rate. For this, the inclusion of PITs, connected to the annular space of the wellbore are another inclusion for the hardware for this project, enabling the communication and integration of the annular pressure into the control logic proposed.

To have a different annular pressure for each individual well, considering this premise, the use of valves as choke devices is necessary, and their actuation will be determined by the pressure that is required by each well. The valves that have been selected for this purpose are designed to control flow between 7 MSCF per day and 220 MSCF per day with a close to linear response for its valve flow coefficient (C_v). These valves are equipped with an electric actuator which will be connected to a SCADA system to be controlled remotely and enabling automation capabilities.

Through IoT, an edge computer device enables the integration of signals and the control of different actuators, utilizing data from different sources, including field information coming from ESPs, PITs, the compressor system and several low-frequency data sources, such as the production data management system (PDMS) in use by the Asset. The monitoring and surveillance of this information and equipments will be managed by the existing Asset Integrated Management Center (AIM) (Yanez et al., 2024).

Software

The main goal of the project is to avoid or reduce gas handling problems in the ESP and its associated production losses by incorporating hardware and software that enables remote and autonomous control over the annular pressure for each well. To determine automatically the deviations from an optimal gas handling response from the ESP, the real time operational parameters for each well are taken into consideration. Specifically motor current response, and its variations in an erratic fashion over its operational mean are a key indicator of gas impeded lift for ESPs (Awaid et al., 2014). For this specific use scenario, the determination of motor current variations considers the absolute average deviation of this parameter over a period of 10 minutes and is the trigger for the control algorithm, meaning that an increase in this parameter over the pre-determined stable conditions will result in the start of the calculation process, this relationship is given by the following equation:

$$Tr = \left| \frac{Dev_{curr_{sc}} - Dev_{curr_{avg}}}{Dev_{curr_{sc}}} \right|$$

Equation 1—Algorithm trigger calculation. Tr = trigger parameter, $Dev_{curr_{sc}}$ = absolute motor current deviation at stable conditions, $Dev_{curr_{avg}}$ = absolute motor current deviation.

The control algorithm starts when $Tr > 1$, and takes into consideration the absolute motor current deviation $Dev_{curr_{sc}}$, which is determined at stable conditions, and $Dev_{curr_{avg}}$ which is the motor current deviation calculated over a predetermined period of 10 minutes.

Once the determination that there is gas impeded lifting for the ESP, the next step is the estimation of pump submergence and the fractions of gas and liquid for the column that submerges the ESP. For this process, the relationships between annular pressure, pump submergence and gas-liquid fractions are in accordance with the empirical relationships established in (McCoy et al., 1988). The main hypothesis is that when the ESP exhibits current fluctuations due to gas impeded lift, there is also a change in conditions for the gas-liquid fraction in the column over the intake of the pump and in order to mitigate these fluctuations, actions need to be taken to go back to stable conditions and thus, a change back to the stable gas-liquid fraction. To evaluate whether the corrective actions were successful, the reference point to get to is the original gas-

liquid fraction that needs to be calculated and stored so that for each step taken to stabilize the annular pressure, a new calculation of these fractions is needed.

The process for determining the gas-liquid fraction is the following:

1. Pressure build-up for the annular space: The first step would be to close the annular space and enable the annular pressure to build up to a recordable value. For this process, a close time was selected based on the historical behavior for these wells and the relationship dP/dt is to be recorded in (psi/min). Where dP is the pressure build-up value and dt is the elapsed build-up time.
2. Determination of the gas-liquid fraction in the column over the intake of the pump: For this calculation, the relationship between intake pressure, column over intake depth and annular pressure is given by the following equation derived from the work established in (McCoy et al., 1988).

$$D_L = D_f - \frac{PIP - P_{cf}}{g_o \cdot f_o + 2.87081 \cdot 10^{-5} \cdot SG_g \cdot P_{cf}}$$

Equation 2—Determination of depth to gas-liquid level interface. D_L = Depth to liquid – gas interphase, D_f = Depth to pump intake, P_{cf} = annular pressure at flow conditions, g_o = gradient of oil column, f_o = gas liquid fraction, SG_g = gas specific gravity.

The depth to the liquid interface or column over the intake D_L is defined by the relationship between D_f or column to the pump intake, taken in true vertical depth (TVD) in ft, PIP given in psi, annular pressure at flow conditions P_{cf} in psi, g_o , the gradient of oil column in psi/ft, f_o , the gas – liquid fraction and SG_g , the specific gravity of gas. Given that the above equation has an implicit relationship between two unknown values, depth to the column over the intake of the pump, D_L and the gas – liquid fraction for said column, f_o , an iterative process is needed to obtain both the depth to the column, and the gas – liquid fraction, for that, a calculation of an adjusted depth to the column, D_a is needed:

$$D_a = D_L + (1 - f_o)(D_f - D_L)$$

Equation 3—Adjusted depth to the liquid-column, D_a .

Taking into consideration the pressure buildup in the annular and this adjusted depth to the liquid column, the empirical relationship presented in (McCoy et al., 1988), between $D_a \cdot \frac{dP}{dt}$ and f_o is given in Fig. 6:

Using the relationship given in Fig. 6, the iterative process starts with an initial value of $f_o = 1$ and then proceeds with the calculation of D_L as given before until stable values for both D_L and f_o are found.

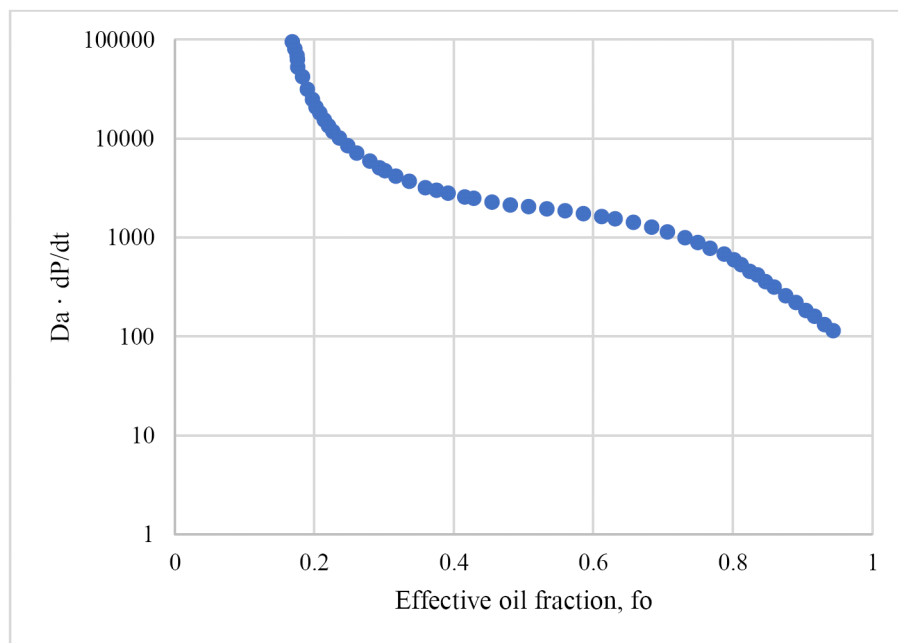


Figure 6—Relationship between effective oil fraction in the column over the intake and gaseous liquid column gradient correction (McCoy et al. 1988).

For this case, stable conditions for the majority of the wells in the well pad were determined when the flare was in operation and for other cases a period of operational stability was analyzed while producing without the flare in operation. Using edge and cloud computing, the liquid fraction was determined for each well and the next step was to modify the annular pressure by opening or closing the control valves through IoT enablers. For this process, the valve is opened or closed in discrete steps and at each step, motor current deviations and calculations for gas – liquid fractions and column depths are made until stability is reached in terms of motor current variations (Fig. 7)

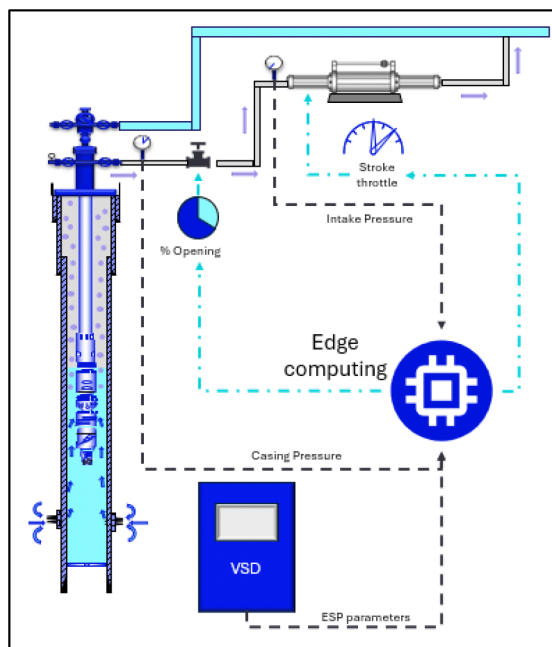


Figure 7—Implementation schematic.

One limiting factor considered for this implementation is that at all times there should be a minimum submergence established for each well, so to avoid any ESP and equipment integrity issues.

Visualization

In practical terms, the project incorporates data from different sources including PDMS, real time high frequency data from VSDs, and real time high frequency data from surface instrumentation. This data feeds the control loops as described above, both for the activation and opening of the valves as well as the control loops for the compressor stroke rate. This system is also configured to the existing SCADA infrastructure to be able to have manual control overrides over the opening of the valves. As for the visualization and monitoring capabilities, all the parameters are monitored in a cloud interactive environment, giving insights in real time of the operating conditions remotely (Fig. 8).



Figure 8—Variables visualization and monitoring.

Results

The well pad where this solution was implemented had a permanent flare onsite that enabled stable production (Table 1) from the ESP lifted wells. The mass of CO_{2e} is calculated with a basis of a 98% combustion efficiency, 47% of CH₄ content and considering a factor of 0.055 kg of CO₂ per MSCF of CH₄ (American Petroleum Institute, 2021).

Table 1—Average daily production parameters and emissions for 2024 with a permanent flare onsite.

Daily average gas production (SCFD)	1,001,829.86
Average daily oil production (BOPD)	7,451.4
kg of CO _{2e} after flaring average daily	25,579.15
Emissions intensity (kgCO _{2e} /BOE)	3.43

The emissions intensity of 3.43 kgCO_{2e}/BOE considers that all the associated produced gas is flared onsite, based on the average daily related production during calendar year 2024, prior to this project. After the implementation of the annular gas management solution, by incorporating the compressor to send annular gas to the flowline and to the central processing station, onsite flaring activities reduced to a minimum with the only recordable flaring related to compressor downtime for maintenance or due to

unforeseen field operational conditions to not affect production stability. The results are presented in [Table 2](#) and consider production affected by depletion after the deployment of the project and take into account the limited days where flare onsite was required.

Table 2—Average daily production parameters and emissions for the implementation period of this solution.

Daily average gas production (SCFD)	825,450.02
Average daily oil production (BOPD)	5,513.4
Total kg of CO _{2e} during compressor downtime	400,439.15
kg of CO _{2e} after flaring average daily	628,527.6
Emissions intensity onsite (kgCO _{2e} /BOE)	0.64

As expected, onsite flaring-related emissions reduced significantly, greatly improving the flaring intensity for that specific well pad. Even though compressor downtime related emissions are considered, emissions intensity reduced 5-fold, with the added benefit of sending the gas to a central processing station where it could be utilized as an energy source.

The production losses that can be addressed with this project are related to gas handling issues ([Fig. 9](#)) such as gas blockages of ESPs and inefficient lift. As shown in [Fig. 10](#), once the flare was removed, the annular gas management system was installed, allowing the wells to operate in stable and optimized conditions, and when the compressor is shutdown, the annular space gets pressurized, generating an increase in the PIP value by 50 psi in average, resulting in production losses due to a production restriction. Based on the results, this innovative solution has allowed the closure of the well pad flare, while minimizing the production losses related to gas handling issues and optimizing the production performance in an average of 130 BOPD for the well pad.

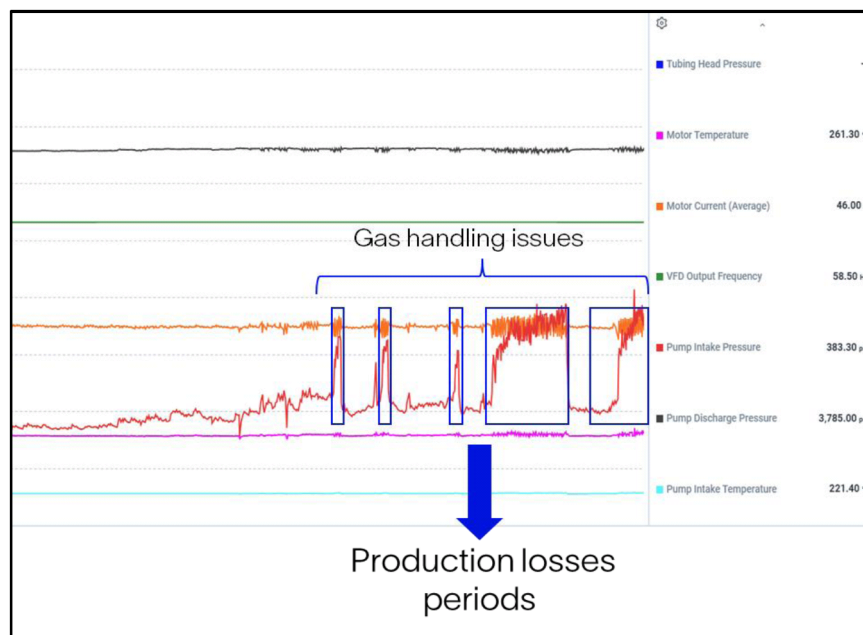


Figure 9—Gas handling events.



Figure 10—PIP increment due to compressor downtime.

The enhancement in production stability as previously stated results in an improvement in the expected RL of the equipment and thus in the reduction of deferred production due to waiting for intervention times. Over the lifetime of the field, cases of fatigue related failures can reduce the RL of the equipment if lifting conditions are not properly managed and gas related lift inefficiencies are not mitigated. An example of such conditions can be seen in Fig. 11, where continuous operation under high current fluctuations is the norm and where these instabilities result in a low RL of 120 days. This can be compared with Fig. 12 where for the same well, a more stable production profile results in a markedly improved RL of 420 days in the ESP. Therefore, by allowing these stable conditions, this solution indirectly improves the expected RL of the equipment.

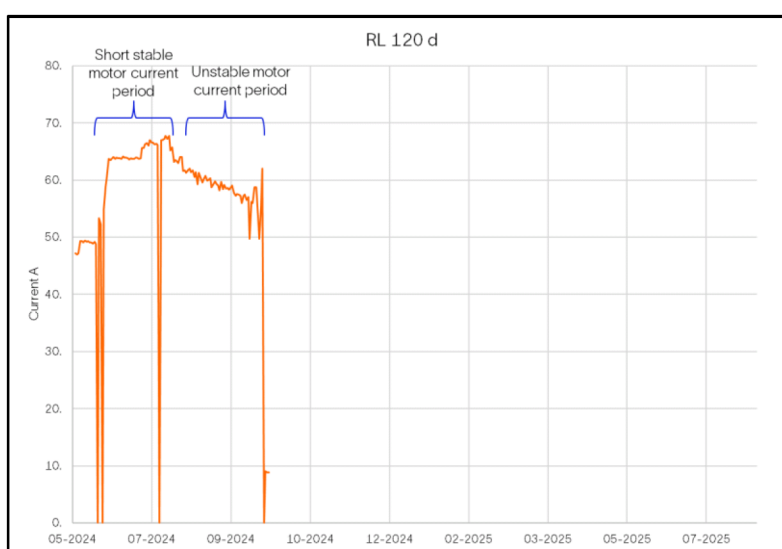


Figure 11—Motor current profile for a reduced RL.

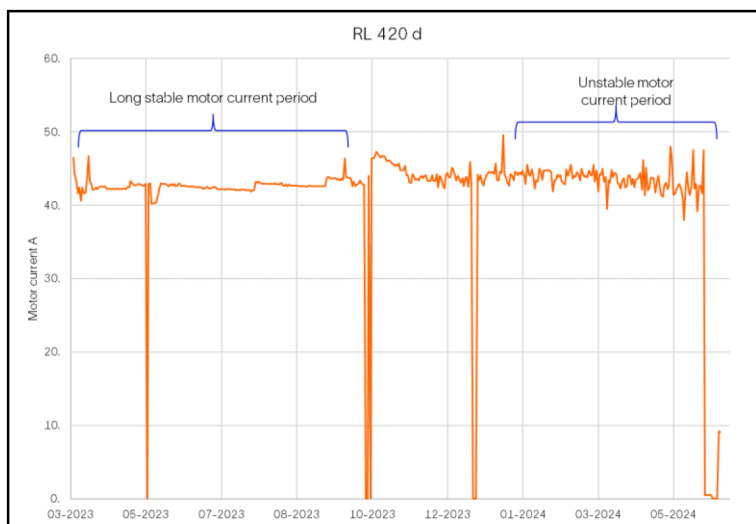


Figure 12—Longer RL correlated with a longer stable current period.

Conclusions

The successful implementation of the SPO annular gas management solution has resulted in no-flaring production operations for the most productive well pad in the field, reducing significantly GHG emissions by 5 times, resulting in an emissions intensity of 0.64 kgCO₂e/BOE. Thus, this project directly contributes to lower the emissions intensity of the Asset, considering that this well pad produces more than 10 percent of the field.

The main goal of this project was reached due to the elimination of 24/7 flaring operations while not compromising production stability. This action supported the reduction of the emissions intensity to be as low as possible, with the only recordable flaring being related to compressor downtime for maintenance or due to unforeseen field operational conditions to not affect production stability and generate production losses.

By implementing a solution utilizing IoT enabled devices and edge and cloud computing, an improvement in production operations was reached, while at the same time supporting the operator to comply with local regulations regarding flaring on-site. This solution is a clear example of employing digital-enabled technology to keep sustained production and improve it by minimizing the effects of gas handling issues due to the increase in annular pressure resulting in higher PIP and thus a production restriction.

This solution employs physics and empirical based correlations by utilizing the long-established correlation between annular pressures, pump submergence and the make-up of the column over the intake of the ESP to do real time optimization of pump submergence. On the other hand, a key benefit is the enabling of remote and autonomous control over a previously manual-only controlled parameter, the annular pressure, and utilized it to improve the lifting efficiency of the ESP by reducing events where motor current variations are due to a direct effect of gas management problems.

As current variations related to gas handling problems with ESPs are minimized, the RL of the equipment gets significantly improved by the stability of the operational parameters and in turn will result in minimizing deferred production due to waiting for intervention times, aiding in reserves extraction and reducing operational costs.

The implementation of this solution, providing control over a previously untapped variable, the well's annular pressure, is a significant step in reaching autonomous operations, a game changer that is taking center stage in Ecuador, marking this development not only a contribution to the client's objectives, but pushing the Ecuadorian oil & gas industry to the forefront of smart, intelligent and sustainable production operations.

Acknowledgements

The authors would like to thank the Operations, Production Engineering and Facilities teams at Consorcio Shushufindi (CSSFD), Tecpetrol and SLB for their unwavering support in the successful implementation of this project and helping advance SPO in a brownfield, and for their support and permission to publish this paper. This work could not have been completed without the multidisciplinary collaboration that the team provided.

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